

Output: True
Input: x = 10001, y = 10
Output: False

Sample Input

Sample Output

625 5	True
-------	------

Sample Input

Sample Output

128 5	False
-------	-------

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Section 2 - ZOHO 1

Section Summary

- No. of Questions: 125
- Duration: 100 min

Additional Instructions:
None

Q1. 1. Find the output for the following programs
#include<stdio.h>
Void main()
{
int i;
for(i = 1 ; i < 4 ; i++)
{
switch(i)
{
case 1 : printf("%d" , i);break;
case 2 : printf("%d" , i);break;
case 3 : printf("%d" , i);break;
}
}
switch(i)
{
case 4 : printf("%d" , i);break;
}
} _____

Q2. 1. Find the output
void main()
{
int a[10] , i = 0;
for(i = 0 ; i<10 ; i++)
a[i] = 9 - i;
for(i = 0 ; i < 10 ; i++)
a[i] = a[a[i]];
for(i = 0 ; i < 10 ; i++)
printf("%d " , a[i]);
} _____

Q3. 1. Find the output
void printout(char* pstr)
{
int iretval = 0;
if(pstr)
{
while(*pstr && *pstr<= '9' && *pstr >= '0')
{
iretval = (iretval*10)+(*pstr - '0');
pstr++;
}
}
printf("%d\n" , iretval);
}
void main()
{
printout("X32");
printout("47X74");
} _____

Q4. 1. Find the output
void main()
{
char *s = "\12345s\n";



```
printf("%d" , sizeof(s));
} _____
```

Q5. 1. Find the output

```
void main()
{
    int a[] = { 8 , 9 , 9 , 9};
    int r[5] = { 0 , 0 , 0 , 0 , 0};
    int i = 0 , m = 1 , s = 4;
    for( i = s-1 ; i >= 0 ; i--)
    {
        r[i+1] = (a[i] + m ) % 10;
        m = (a[i] + m) /10;
    }
    r[0] = m ;
    for( i = 1 ; i <= s ; i++)
        printf("%d" , r[i]);
} _____
```

Q6. 1. Find the output

```
int i = 0 ;
int fun(int a)
{
    i++;
    if( a > 99)
        return a-12;
    return fun(fun(a+25));
}
void main()
{
    printf("%d " , fun(69));
    printf("%d" , i);
} _____
```

Q7. 1. Find the output ons)

```
int max(int x , int y)
{
    return (y > x) ? y : x;
}
void main()
{
    int a[] = { -6 , - 7 , 8 , - 9 , -2 , 3,-4,5};
    int value = a[0] , ctval = a[0];
    int i = 0 , n = 8;
    for( i = 1 ; i < n ; i++)
    {
        ctval = max( a[i] , ctval+a[i]);
        value = max(value , ctval);
    }
    printf("%d" ,value) ;
} _____
```

Q8. 1. Find the output

```
void main()
{
    int a=30,b=40,x;
    x=(a=10)&(b=50);
    printf("x=%d\n",x);
} _____
```

Q9. 1. Find the output

```
int main()
{
    char *s[]={ "dharmr'a","hewlett-packard","siemens","ibm"};
    char **p;
    p = s ;

    printf("%s" ,++*p);
    printf("%s",*p++);;
    printf("%s" ,++*p);
} _____
```

Q10. 1. Find the output

```
void main()
{
    int x=0,y=1;
    y=x;
    x=!y;
    printf("x=%d y=%d\n",x,y);
} _____
```

Q11. 1. Find the output

```
void main()
{
    int x=3,y=4,z=4;
```

```
printf("ans=%d\n",(z>=y>=x?100:200));
} _____
```

Q12. 1. Find the output

```
void main()
{
    float a=12.25,b=13.65;
    if(a==b)
        printf("a and b are equal");
    else
        printf("a and b are not equal");
} _____
```

Q13. 1. Find the output

```
void main()
{
    int arr[]={0,1,2,3,4};
    int *ptr,i;
    for(ptr=arr+4;ptr >= arr ; ptr--)
        printf("%d ",*ptr);
} _____
```

Q14. Find the output

```
void junk(int,int*);
int main()
{
    int i=-5,j=-2;
    junk(i,&j);
    printf("i=%d j=%d" ,i ,j);

    return 0;
}
void junk(int i,int *j)
{
    i=i*1;
    *j=*j * i;
} _____
```

Q15. Find the output

```
void main()
{
    if('Z' < 'z')
        printf("Pilots are on strike...\n");
    else
        printf("for absolutely outlandish demands\n");
} _____
```

Q16. Find the output

```
void main()
{
    float a = 0.7;
    if( a < 0.7)
        printf("Stoned");
    else
        printf("Avenged");
}
_____
```

Q17. Find the output

```
void main()
{
    float a=0.5,b=0.9;
    if(a&& b>0.9)
        printf("tce-cse-a\n");
    else
        printf("tce-cse-b\n");
} _____
```

Q18. 1. Find the output(branching and looping)

```
void main()
{
    int i;
    for(i=1;i++<=5;printf("%d ",i));
}
_____
```

Q19. Find the output

```
void main()
{
    int i = 1 , j = 1;
    for(;j;printf("%d %d\n",i,j))
        j=i++<=5;
```



```
}  
_____
```

Q20. Find the output

```
void main()  
{  
    int x=3,y,z;  
    z=y=x;  
    z*=y=x*x;  
    printf("x=%d y=%d z=%d\n",x,y,z);  
} _____
```

Q21. Find the output

```
void main()  
{  
    int x=3,z;  
    z=x/++x;  
    printf("x=%d z=%d\n",x,z);  
}  
_____
```

Q22. Find the output

```
void main()  
{  
    int x , y , z;  
    x=y=z=1;  
    z=++x || ++y&&++z;  
    printf("x=%d y=%d z=%d \n",x,y,z);  
}  
_____
```

Q23. Find the output

```
int main()  
{  
    char *s[]={ "dharmr'a","hewlett-packard","siemens","ibm"};  
    char **p;  
    p = s ;  
  
    printf("%s",++*p);  
    printf("%s",*p++); ;  
    printf("%s",++*p);  
}  
Output: _____
```

Q24. Find the output

```
void main()  
{  
    char ch='E';  
    switch(ch)  
    {  
        case(ch>=65 && ch<=90):  
            printf("Capital letter\n");  
            break;  
        case(ch>=97 && ch<=122):  
            printf("small letter\n");  
            break;  
        case (ch>=48 && ch<=57):  
            printf("Digit");  
            break;  
        default:printf("Anyother");  
    }  
}  
_____
```

Q25. Find the output

```
void main()  
{  
    int i = 3;  
    switch(i)  
    {  
        case 1: printf("cse\t");  
        case 2: printf("It\n");break;  
        case 3: continue;  
        default : printf("goodbye");  
    }  
}  
_____
```

Q26. Find the output

```
void main()  
{  
    char s;  
    switch(s)  
    {  
        case '1': printf("database");  
    }
```



```
        case '2': printf("data-structure");
        default: printf("c");
        printf("byebye");
    }
}
```

Q27. Find the output

```
void main()
{
    int k=-2,j=4;
    switch(k/=j/k)
    {
        default:printf("lenovo");
        case 0 : printf("hp");
        case 1: printf("acer");
        case 2: printf("dell");
    }
}
```

Q28. Find the output

```
void main()
{
    int j,x=0;
    for (j=0;j<=5;j++)
    {
        switch(j-1)
        {
            case 0:
            case -1:
                x -= 1;break;
            case 1:
            case 2:
            case 3:
                break;
            default: x+=3;
        }
        printf("%d ", x);
    }
}
```

Q29. Find the output

```
void main()
{
    int i;
    for(i = 2 ; i <= 10 ; i++)
    {
        switch(i)
        {
            case 2: printf("0");continue;
            case 3: break;
            case 4:
            case 5:printf("1");break;
            default: printf("000");
        }
    }
}
```

Q30. Find the output

```
void main()
{
    char ch='E';
    switch(ch) {
    case(ch>=65 &&ch<=90):
        printf("Capital letter\n");
        break;
    case(ch>=97 &&ch<=122):
        printf("small letter\n");
        break;
    case (ch>=48&&ch<=57):
        printf("Digit");
        break;
    default: printf("Any other");
    }
}
```

Q31. Find the output

```
#include<stdio.h>
#include<malloc.h>
#include<string.h>
int main()
{
    int i;
```

```
char a[]="String";
char *p = "New String";
char *temp;
temp = malloc(strlen(p) + 1);
p = malloc( strlen(temp) + 1);
strcpy(p , temp);
printf("%s" , p);
}
Output : _____
```

Q32. Find the output

```
int funcl(int k)
{
    k++;
    return k;
}

void main()
{
    int k=35,z;
    k=funcl(k=funcl(k=funcl(k)));
    printf("k =%d\n",k);
}
_____
```

Q33. Find the output

```
int main()
{
    int n = 12 , res = 1;
    while( n > 3)
    {
        n -= 3;
        res *= 3;
    }
    printf("%d" , n*res);
}
Output : _____
```

Q34. Find the output

```
void fun(int b[][3]);
int main()
{
    int a[3][3] = {9,8,7,6,5,4,3,2,1};
    fun(a);
    printf("%d\n" , a[2][1]);
}
void fun(int b[][3])
{
    ++b;
    b[1][1]=5;
}
Output : _____
```

Q35. Find the output

```
void pri(int,int);
void printit(float,int);
void main()
{
    float a=3.14;
    int i=99;
    pri(i,a);
    printit(a,i);
}

void pri(int i,int a)
{
    printf("i=%d a=%f\n",i,a);
    printf("a=%f i=%d\n",a,i);
}
void printit(float a,int i)
{
    printf("a=%f i=%d\n",a,i);
    printf("i=%d a=%f\n",i,a);
}

_____
_____
_____
_____
```

Q36. Find the output

```
void main()
{

    int i , n;
    char x[5];
    strcpy( x , "Zoho");
    n = strlen(x);
    *x = *(x+(n-1));
    printf("%s" , x);
}
```


Output: _____

Q37.

Find the output
void main()
{
 int c[]={5,4,3,4,5};
 int j , *q = c;
 for(j = 0 ; j<5 ; j++){
 printf("%d" , *c);
 ++q;
 }
}

Output: _____

Q38.

Find the output
void main(){
 int k=35,*z,*y;
 z=&k; y=z;
 *z++=*y++;
 k++;
 printf("k=%d z=%d y=%d",k,z,y);
}

Q39.

Find the output
void main()
{
 int i = 1;
 for(i =0 ; i= -1 ; i=1){
 printf("%d", i);
 if(i!= 1) break;
 }
}

Output: _____

Q40.

Find the output
void main()
{

 int s[] = {1,0,5,0,10,0};
 int f[] = {2,4,6,8,10,12};
 int n = 6 , i = 0 , j = 0;
 for(j = 1 ; j < n ; j++)
 {
 if(s[j] >= f[i])
 {
 printf("%d" , i);
 i = j;
 }
 }
}

output : _____

Q41.

Find the output
void main(){
 int a=100,*b,**c,***d;
 b=&a; c=&b; d=&c;
 printf("%d %d %d %d",a,*b,**c,***d);

}

Q42.

Find the output
void main(){
 int z=4;
 printf("%d\n",printf("%d%d\n",z, z-1));
}

Q43.

Find the output
void f(int *a , int m)
{
 int j = 0;
 for(j = 0 ; j < m ; j++)
 {
 *(a+j) = *(a+j) - 5;
 }
}
void main()
{
 int a[] = {'f' , 'g' , 'h' , 'i' , 'j'};
 int j = 0 ;
 f(a , 5);
 for(j = 0 ; j<= 4 ; j++)
 printf("%c\t" , a[j]);
}



```
}
Output: _____
```

Q44. Find the output

```
float *jam(float *r){

r=r+1;
return (r);
}

void main()
{
float *jam(float *);
float p=23.5,*q;
q=&p;
printf("q before call=%d\n",q);
q=jam(&p);
printf("q after call=%d",q);
}

_____
_____
```

Q45. Find the output

```
void main()
{
int i;
printf("hai");
for(i = 1 ; i<= 10 ; i++)
main();

}

_____
```

Q46. Find the output

```
void main()
{
int i=0,j=0 , sum=0;
for(i= 1; i < 500 ; i*=3)
for(j=0;j<i;j++)
sum++;
printf("%d",sum);

}
Output: _____
```

Q47. Find the output

```
void main()
{
if(printf("C for yourself how it works\n"))
main();

}

_____
```

Q48. Find the output

```
void main()
{
int n;
for(n = 6 ; n!= 1; n--)
printf("%d" , n--);
}
Output: _____
```

Q49. Find the output

```
void main()
{
unsigned int ch=0;

for(ch=65;ch<=255;)
printf("%d %c\n",ch,ch++);

}

_____
```

Q50. Find the output

```
void main()
{
int a[3][4] = {2,4,6,5,10,12,12,10,5,6,4,2};
int i = 0 , j , k =99;
while(i < 3)
{
for(j = 0 ; j < 4 ; j= j++)
{
if( a[i][j] < k)
{
```



```
        k = a[i][j];
    }
}
i++;
}
printf("%d" , k);
}
Output : _____
```

Q51. Find the output

```
void main()
{
float a=0.7;
double b=0.7;
if(a==b)
    printf("condition statisfied");
else
    printf("condition not statisfied");

printf("\na=%f b=%lf\n",a,b);
}

_____
_____
```

Q52. Find the output

```
void main()
{

char *x="Alice";
int i , n = strlen(x);
*x = x[n];
for(i=0; i<=n; i++)
{
printf("%s ", x); x++;
printf("\n", x);
}

return 0;

}
Output : _____
```

Q53. Find the output

```
void main()
{
float y=0.9;
long double z=0.9;

if(y==z)
printf("icecrearm");
else
printf("cake");

}

_____
```

Q54. Find the output

```
void change()
{
    auto int i=100;
    register int j=200;
printf("change's i and j are %d %d\n",i,j);

}
void main()
{
auto int i=10;
register int j=20;
printf("main's l and j are %d %d\n",i,j);
change();
printf("main's l and j are %d %d\n",i,j);
}

_____
_____
_____
```

Q55. Find the output

```
void main()
{
double x,d=4.4;
int i=2,y;
x=(y=d/i)*2;
printf("x=%lf y=%d\n",x,y);
y=(x=d/i)*2;
printf("x=%lf y=%d\n",x,y);
}

_____
_____
```



Q56. Find the output

```
void main()
{
double x,d=5.0;
int y;
x=d*(x=2.5/d);
printf("x=%lf\n",x);
x=d*(y=(int)2.5+1.5);
printf("x=%lf y=%d\n",x,y);
}
```

Q57. Find the output

```
struct value{
int bit1:1;
int bit3:4;
int bit4:4;
}bit;
int main()
{
printf("%d\n", sizeof(bit));
return 0;
}
```

Output : _____

Q58. Find the output

```
void main()
{
double x,d=5.0;
int y;
x=d*(x=2.5/d);
printf("x=%lf\n",x);
x=d*(y=(int)2.5+1.5);
printf("x=%lf y=%d\n",x,y);
}
```

Q59. Find the output

```
struct node
{
int data;
float d;
struct node *link;
};
int main()
{
struct node *p, *q;
p = (struct node *) malloc(sizeof(struct node));
q = (struct node *) malloc(sizeof(struct node));
printf("%d, %d\n", sizeof(p), sizeof(q));
return 0;
}
```

Output : _____

Q60. Find the output

```
void main()
{
int c=5;
printf("c=%d\n",c--);
if(c) main();
}
```

Q61. Find the output

```
int func(int x)
{
static int v=2;
v--;
return (v-x);
}
int i;
void main()
{
int j;
for(;;)
{
if( j= func(i) )
printf("j= %d ", j);
else
break;
}
}
```

Q62. Find the output
void main()
{
long num=2;
short n=2;
signed no=2;
printf("num=%ld n=%d no=%d\n",num,n,no);
}

Q63. Find the output
void main()
{
char ch=122,ch1='z';
printf("ch=%c\n",ch);
printf("chl=%d\n",ch1);
}

Q64. Find the output
typedef union
{
int a;
char b[10];
float c;
}Union;
int main()
{
Union x , y = {100};
x.a = 50;
strcpy(x.b , "Hello");
x.c = 21.50;
printf("%d %s %f\n" , x.a , x.b , x.c);
printf("%d %s %f" , y.a,y.b, y.c);
}
Output: _____

Q65. Find the output
void main()
{
unsigned int a=25;
unsigned b=25;
long unsigned c=345L;
long signed d=345L;
printf("a=%u b=%u\n",a,b);
printf("c=%lu d=%d\n",c,d);
}

Q66. Find the output
struct point{

int x;
int y ;
};
struct point origin , *pp;
int main()
{
pp = &origin;
printf("origin is (%d %d)\n" , (*pp).x , (*pp).y);
printf("origin is (%d %d)" , pp->x , pp->y);
return 0;
}
Output : _____

Q67. Find the output
void main()
{
int i = -1;
printf("i =%d +i = %d\n" , i , +1);
}
Output : _____

Q68. Find the output
void main()
{
char not;
not=12;
printf("%d",not);
}
Output : _____



- Q69.

Find the output

```
void main()
{
    auto int i=100;
    printf("i=%d\n",i);
    i+=1;
    printf("i=%d\n",i);
}
_____
_____
```
- Q70.

Find the output

```
#p1
void main()
{
    register int i;
    for(i=1;i<=100;i++)
        printf("%d\n",i);
}
#p2
void main()
{
    auto int i;
    for(i=1;i<=100;i++)
        printf("%d\n",i);
}
_____
```
- Q71.

Find the output

```
#define FALSE -1
#define TRUE 1
#define NULL 0
void main()
{
    if(NULL)
        puts("NULL");
    else if(FALSE)
        puts("TRUE");
    else
        puts(" FALSE");
}
Output : _____
```
- Q72.

Find the output

```
#define CUBE(x) x*x*x
void main()
{
    int a;
    a= 27 / CUBE(3);
    printf("%d" , a);
}
_____
```
- Q73.

Find the output

```
#define CUBE(x) (x*x*x)
void main()
{
    int a , b;
    a = CUBE(b+4) / b++;
    printf("a= %d b = %d ", a , b);
}
_____
```
- Q74.

Find the output

```
void main()
{
    int k = 1;
    printf("%d==1 is "" %s" ,k, k == 1 ? "TRUE":"FALSE");
}
Output : _____
```
- Q75.

Find the output

```
#define AND &&
#define OR ||
#define LE <=
#define GE >=
void main()
{
    char ch='D';
    if((ch GE 65 AND ch LE 90) OR (ch GE 97 AND ch LE 122))
        printf("Alphabet\n");
}
```



```
else
    printf("Not alnhabet");

}
```

Q76. Find the output

```
void main()
{
static float arr[]={1.2,12,2.4,24,3.5,35};
int i;
for(i=0;i<=5;i++)
printf("%f ",arr[i]);
}
```

Q77. Find the output

```
void main()
{
static int b[]={10,20,30,40,50};
int i;
for(i = 0; i<= 4 ; i++)
printf("%d ",b[i]);
}
```

Q78. Find the output

```
int main()
{
int t , i ;
for ( t=4;scanf("%d",&i)-t;printf("%d\n",i))
printf("%d--",t--);
}
```

Output : _____

Q79. Find the output

```
void main()
{
static int a[5]={5,10,15,20,25};
int i,j,m,n;
i=4-a[1];
j=a[1]++;
printf("i=%d j=%d a[1]=%d\n",i,j,a[1]);
i=1;
m=a[1]+41;
printf("i =64 m=%d\n",i,m);
i=2;
n=a[1]++;
printf("i=%d n=%d\n",i,n);
}
```

Q80. Find the output

```
struct emp{
int len;
char name[1];
};
int main()
{
char newname[] = "Rahul";
struct emp *p = (struct emp *) malloc(sizeof(struct emp) -1 + strlen(newname)+ 1);
p->len = strlen(newname);
strcpy(p -> name, newname);
printf("%d %s\n", p->len, p->name); return 0;
}
```

Output : _____

Q81. Find the output

```
void main()
{
static int a[]={10,20,30,40,50};
int j;
for (j=0;j<5;j++)
{
printf("%cl\n",*a);
a++;
}
}
```

Q82. Find the output

```
int main() {
printf("%d %d %d %d\n",72,072,0x72,0X72);
}
```



```
return 0;
}
Output : _____
```

Q83. Find the output

```
void main()
{
    char ch;
    int a;
    float b;
    printf("bytes occupied by ch=%d\n",sizeof(ch));
    printf("bytes occupied by a=%d\n",sizeof(a));
    printf("bytes occupied by b=%d\n",sizeof(b));
}
Output :
_____
_____
_____
```

Q84. Find the output

```
void main()
{
    printf("%d\n" , sizeof('7'));
    printf("%d\n" , sizeof(7));
    printf("%d\n" , sizeof(7.0));
}
Output: _____
_____
_____
```

Q85. Find the output

```
void main()
{
    char ch=291;
    printf("%d %d %c\n",2147483648,ch,ch);
    return 0;
}
Output : _____
```

Q86. Find the output

```
void main()
{
    static int b[]={10,20,30,40,50};
    int i,*k;
    k= &b[4]-4;
    for(i=0;i<=4;i++)
    printf("%d ",*k);
    k++;

}
_____
```

Q87. Find the output

```
void main()
{
    int g;
    g=300000*300000/300000;
    printf("g=%d\n",g);
}
Output : _____
```

Q88. Find the output

```
void main()
{
    static int a[]={2,4,6,8,10};
    int i;
    for(i=0;i<=4;i++)
    *(a+i)=a[i]+i[a];
    printf("%d\n",*(i+a));

}
_____
```

Q89. Find the output

```
void main()
{
    float a;
    a=4/2;
    printf("%f %f\n",a,4/2);

}
Output : _____
```



- Q90.

Find the output

```
void main()
{
    int arr[]={0,1,2,3,4};
    int i,*ptr;
    for(ptr=&arr[0],i=0;i<=4;i++)
        printf("%d ",ptr[i]);
}
```

- Q91.

Find the output

```
void main()
{
    printf("%d\n",sizeof(4)/sizeof(2.0));
    printf("%d\n",sizeof(2.0)/sizeof(4));
}
```

Output : _____
- Q92.

Find the output

```
void main()
{
    int arr[]={0,1,2,3,4};
    int i,*p;
    for(p=arr,i=0;p+i<=arr+4;p++,i++)
        printf("%d ",*(p+i));
}
```

- Q93.

Find the output

```
void main()
{
    int x=10,y=5,p,q;
    p=x > 9;
    q=x>3&& y!=3;
    printf("p=%d q=%d \n",p,q);
}
```

Output : _____
- Q94.

Find the output

```
void main()
{
    static int a[]={0,1,2,3,4};
    static int *p[]={a,a+1,a+2,a+3,a+4};
    int **ptr=p;
    printf("%d %d\n",a,*a);
    printf("%d %d %d\n",p,*p,**p);
    printf("%d %d %d\n",ptr,*ptr,**ptr);
}
```

- Q95.

Find the output

```
void main()
{
    static int a[]={0,1,2,3,4};
    static int *p[]={a,a+1,a+2,a+3,a+4};
    int **ptr=p;
    printf("%d %d %d\n",ptr-p,*ptr-a,**ptr);
    *ptr++;
    printf("%d %d %d\n",ptr-p,*ptr-a,**ptr);
    *++ptr;
    printf("%d %d %d\n",ptr-p,*ptr-a,**ptr);
    ++*ptr;
    printf("%d %d %d\n",ptr-p,*ptr-a,**ptr);
}
```

- Q96.

Find the output

```
void main()
{
    static int n[3][3]={12,4,3,6,8,5,3,5,11};
    printf("%d %d %d\n",n,n[2],n[2][2]);
}
```

- Q97.

Find the output

```
void main()
{
    char s[]="Rendezvous !";
    printf("%d\n",*(s+strlen(s)));
}
```



```
}
_____
```

Q98. Find the output

```
void main()
{
char str[20];
static int i;
for(;;) {
    i++[str]='A'+2;
    if(i==19)
        break;
}
i[str]=0;
printf("%s" , str);

}
_____
```

Q99. 1. Find the output(strings)

```
void main()
{
char s[]="C smart!!";
int i;
for(i=0;s[i];i++)
printf("%c%c%c%c%c\n",s[i],*(s+i),i[s],*(i+s));

}
Output :
_____
```

Q100. 1. Find the output

```
void main()
{
char s[]="Dinks Grunts and Guffaws";
printf("%c\n",*(&s[2]));
printf("%s\n",(s+5));
printf("%s\n",s);
printf("%c\n",*(s+10));
}

_____
_____
_____
_____
```

Q101. Find the output

```
void main()
{
    int a = 3 , b = 2 , c = 1 , d;
    d = a | b & c;
    printf("d = %d\n", d);
    d = a+ b & -c;
    printf("d = %d\n" , d);
}

_____
```

Q102. Find the output

```
void main()
{
char str[]="MalayalaM";
char *s;
s = str+8;
while( s > str)
{
    printf("%c" , *s);
    s--;
}
}

_____
```

Q103. Find the output

```
int show();
void main()
{

    int (*f)();
    f= show;
    printf("address= %d\n",f);
}
int show()
{
    printf("Diamonds are very costly");
}

_____
```

Q104. Find the output

```
void main()
```




```
{
char str[]="Shall we tell the Deputy Director?";
printf("%s\n%s\n%s\n",str,str+6,str+9);

}

_____
_____
_____
```

Q105.

Find the output

```
struct num
{
    unsigned bit0:1;
    unsigned bit1:1;
    unsigned bit2:1;
    unsigned rest:5;
};
union a
{
    struct num n;
    char ch;
}b;
void main()
{
    b.ch = 32;
    printf("%d %d %d %d", b.n.bit0 , b.n.bit1,b.n.bit2,b.n.rest);
}
```

Q106.

Find the output

```
void main()
{
    printf("%f\n" , (float)(int)(float)(int)6.5/2+3.5);
}
```

Q107.

Find the output

```
struct employee
{
    char name[25];
    int age;
    float bs;
};

void main()
{
    struct employee e;
    e.name = "Hacker";
    e.age=25;
    printf("%s %d" , e.name , e.age);
}
```

Q108.

Find the output

```
typedef struct
{
    char name[20];
    int age;
}a;
void main()
{
    a temp= {"sunil" , 30};
    printf("%s %d" , temp.name , temp.age);
}
```

Q109.

Find the output

```
struct name1
{
    char name[25];
    char lang[10];
};
static struct name1 a = {"Hacker" , "cr"};
void main()
{
    printf("%s %s" , a.name , a.lang);
}
```

Q110.

Find the output

```
void main()
{
    enum status {low,medium,high};
    enum status rain;
    rain = 0;
    if(rain == low)
```



```
printf("rain = %d", rain);
}
```

Q111. Find the output

```
struct a {
    char ch[7];
    char *str;

};

void main()
{
    static struct a s1={"Nagpur" , "Bombay"};
    printf("%c %c\n" , s1.ch[0] , *s1.str);
    printf("%s %s" , s1.ch , s1.str);
}
```

Q112. Find the output

```
void main()
{
    int a=0Xff;
    if(a<<4>>12)
        printf("lefttest");
    else
        printf("rightest");
}
```

Q113. Find the output

```
void main()
{
    int a=0Xff;
    if(a<<4>>12)
        printf("lefttest");
    else
        printf("rightest");
}
```

Q114. Find the output

```
struct a
{
    int i;
    char ch[4];
};
union b
{
    int j;
    char ch[4];
};
void main()
{
    printf("%d " , sizeof(struct a));
    printf("%d " , sizeof(union b));
}
```

Q115. Find the output

```
void main()
{
    short int k;
    k = -35;
    printf("k=%d " , k);
    k = -k;
    printf("k = %d " , k);
}
```

Q116. Find the output

```
int main(int argc , char* argv[])
{
    printf("%d " , argc);
    printf("%s" , argv[0]);
}
```

Q117. Find the output

```
union a
{
    int i;
    char ch[2];
};
void main()
```

```
{
union a u;
u.i = 256;
printf("%d %d %d ", u.i , u.ch[0] , u.ch[1]);
}
_____
```

Q118.

Find the output

```
{
    long int i;
    char ch[4];
};
void main()
{
    struct a s;
    s.i = 512;
    printf("%d %d %d" , s.ch[0] , s.ch[1] ,s.ch[3]);
}
_____
```

Q119.

Find the output

```
void main()
{
    static char str[]="Triplet";
    char *s;
    s = str;
    while(*s)
    {
        putc(*s , stdout);
        fputc(*s);
        printf("%c\n ",*s);
        s++;
    }
}
_____
```

Q120.

Find the output

```
void main()
{
    char name[20]="Sandeep";
    int salary=1500;
    printf("%s %d\n", name , salary);
    fprintf(stdout , "%s%d\n",name,salary);
}
_____
_____
```

Q121.

Find the output

```
{
    int i;
    char ch[4];
};
void main()
{
    union a u;
    u.ch[0]=3;
    u.ch[1]=2;
    u.ch[2]=0;
    u.ch[3]=0;
    printf("%d %d %d",u.ch[0],u.ch[1], u.i);
}
_____
```

Q122.

Find the output

```
void main()
{
    printf("I\\tam\\ta\\tboy\\n");
}
_____
```

Q123.

Find the output

```
void main()
{
    float a=3.14;
    printf("a=%f\n",a);
    printf("a=%6.2f\n",a);
    printf("a=%-6.2\n",a);
    printf("a=%6.1f\n",a);
    printf("a=%6.0f\n",a);
}
_____
_____
_____
_____
_____
```



Q124.

Find the output
void main()
{
printf("Hello\nHi\n");
printf("Hello\rHi\n");
printf("Hello\b\b\b\b\b\n");
printf("Hil\b\b\bBye\n");
}

Q125.

Find the output
void main()
{
printf("%20s\n","short leg");
printf("%20s\n","long leg");
printf("%20s\n","deep fine leg");
printf("%20s\n","backward short leg");
printf("%20s","legs are the same");
}

Section 3 - ZOHO

Section Summary

- No. of Questions: 6
- Duration: 100 min

Additional Instructions:

None

Q1.

New Number System
Form a number system with only 3 and 4. Find the nth number of the number system.
Eg.) The numbers are: 3, 4, 33, 34, 43, 44, 333, 334, 343, 344, 433, 434, 443, 444, 3333, 3334, 3343, 3344, 3433, 3434, 3443, 3444

Input Format

Input will be an integer

Output Format

Print the nth number

Constraints

1<=N<=10000000

Sample Input

Sample Output

10

344

Sample Input

Sample Output

6743

434334344333

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q2.

Word Reversal form the first occurrence of sub-string
Write a program to accept two strings S1 and S2 and reverse the words of S1,starting from the word where the first occurrence of S2 present in S1. Same empty spaces between the words must be maintained in the output. Write the program without splitting up the strings into array of words.
DON'T use any inbuilt functions
Input: S1= This is a test input string S2=st
Output : This is a string input test

Sample Input

Sample Output

this is a test sentence
st

this is a sentence test

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q3.

Permutation of string



A permutation, also called an “arrangement number” or “order,” is a rearrangement of the elements of an ordered list S into a one-to-one correspondence with S itself. A string of length n has n! permutation.
Below are the permutations of string ABC.
ABC ACB BAC BCA CBA CAB

Sample Input

Sample Output

ABC

ABC ACB BAC BCA CBA CAB

Sample Input

Sample Output

1234

1234 1243 1324 1342 1432 1423 2134 2143 2314 2341 2431 2413 3214 3241

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q4. **Locker Problem**
There is a school with 100 students, and correspondingly 100 lockers, all of which start off closed. The first student opens every locker. The second student closes every other locker, starting with the second (2, 4, 6 etc). The third student changes the state of every third locker starting with the third (3,6,9 etc). The fourth would change the status of lockers numbered 4,8,12 etc.. That is, if the locker is open, it is closed, and if it is closed, it is opened. This continues until all 100 students have passed along the lockers. After the 100th student is done, which lockers are open and which are closed?
[Note: program should work for any number of students/lockers]

Sample Input

Sample Output

100

open = 10
close = 90

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q5. **Find the Path**
Given an (m x n) matrix, write a program to traverse the cell and print the values present in the given path. Included necessary validation and proper error messages in case of given path is out of bounds.
5 x 5 matrix :
{1 2 3 4 5 } (row 1)
{6 7 8 9 0 } (row 2)
{1 2 3 4 5 } (row 3)
{6 7 8 9 0 } (row 4)
{1 2 3 4 5 } (row 5)

Path Notation : ">" is going right, "v" going down, "<" is going left, "^" is going up.

Example Input 1 :
Start at (Row, Column): 1, 2
Path: >>> v
Output: 2 3 4 5 0

Example Input 2:
Start at (Row, Column): 2,3
Path: v > > v < < ^ > > v v

Output 8 3 4 5 0 9 8 3 4 5 0 5

Example Input 3:
Start at(row, Column): 1 , 4
Path: > v > >

Output: Invalid Path

Input Format

N M - matrix row and col
input for matrix
startrow startcol
path string

Sample Input

Sample Output

5 5
1 2 3 4 5
6 7 8 9 0
1 2 3 4 5

8 3 4 5 0 9 8 3 4 5 0 5

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q6. **Group anagram words**
Given array of words, group the anagrams and print. Any word or phrase that exactly reproduces the letters in another order is an anagram. Arrive most efficient algorithm.

Examples :
Input: {tar,rat,banana,atr,nanaba}
Output: Anagrams:
rat atr tar
nanaba banana

Input: {abc, cde, xyz, dec}
Output: Anagrams:
cde dec
Others:



Abc
xyz

Input Format

N - no of words
get input words into array

Sample Input

5
tar rat banana atr nanaba

Sample Output

rat atr tar
nanaba banana

Time Limit: - ms Memory Limit: - kb Code Size: - kb

